Setup Guide – Open Keg Scale & Transfer Keg Scale

Both projects use a Wemos D1 Mini (ESP8266) with an HX711 ADC and 4 × 50 kg half-bridge load cells.

Choose the project that fits your needs:

	Open Keg Scale	Transfer Keg Scale
Display	Web dashboard (open-plaato-keg server)	Local OLED 128×64
Protocol	Blynk binary TCP	HTTP REST API
Build tool	Arduino IDE	PlatformIO
Use case	Permanent tap monitoring	Portable / transfer use

Table of Contents

---

1. Hardware & Parts

Shared (both projects)

Part	Notes
Wemos D1 Mini	ESP8266, 4 MB flash — 2.4 GHz WiFi only
HX711 breakout	Any common board; 24-bit ADC
4 × 50 kg half-bridge load cell	RED / BLACK / WHITE 3-wire cells
Micro-USB cable	For initial flash
5 V USB power supply	≥ 500 mA

Transfer Keg Scale only

Part	Notes
OLED 128×64	SSD1306 or SH1106, I2C, 4-pin (VCC / GND / SDA / SCL)

Optional

Part	Notes
Load cell combinator PCB	Simplifies the Wheatstone bridge wiring
3D-printed enclosure	STL files in `open_keg_scale/3d printer files/` (Open Keg Scale), `transfer-keg-scale/transfer scale.3mf` (Transfer)

---

2. Wiring

Both projects share the same load cell → HX711 wiring.

Refer to the `Wiring info.jpg` in each project folder for a visual diagram.

2a. Load Cells → HX711 (Wheatstone Bridge)

Wire the four half-bridge cells as one full bridge.

Each cell has three wires: RED (excitation +), BLACK (excitation −), WHITE (signal).

```
Cell A (front-left) RED ─┐ Cell B (front-right) RED ─┤──→ HX711 E+ Cell C (back-right) RED ─┤ Cell D
(back-left) RED ─┘ Cell A (front-left) BLACK─┐ Cell B (front-right) BLACK─┤──→ HX711 E− Cell C
(back-right) BLACK─┤ Cell D (back-left) BLACK─┘ Cell A (front-left) WHITE─┐ Cell C (back-right)
WHITE─┘──→ HX711 A+ (diagonal pair) Cell B (front-right) WHITE─┐ Cell D (back-left) WHITE─┘──→ HX711
A− (opposite diagonal pair)
```

> *Why diagonals? Opposite platform corners flex the same direction under load. Wiring them to the same signal leg gives additive output and cancels common-mode errors.*

If your kit included a combinator PCB, wire each cell's three leads to it and connect the PCB's four output lines (E+, E−, A+, A−) directly to the HX711.

2b. HX711 → Wemos D1 Mini

Open Keg Scale

HX711	D1 Mini pin	GPIO
VCC	3V3	—
GND	GND	—

HX711	D1 Mini pin	GPIO
DT	D6	12
SCK	D5	14

Transfer Keg Scale

HX711	D1 Mini pin	GPIO
VCC	5V	—
GND	GND	—
DOUT	D5	14
SCK	D6	12

> *The Transfer Keg Scale powers the HX711 from 5 V for better noise immunity.*

2c. OLED → D1 Mini (Transfer Keg Scale only)

OLED	D1 Mini pin	GPIO
VCC	3V3	—
GND	GND	—
SDA	D2	4
SCL	D1	5

2d. Buttons & LED (Open Keg Scale)

Signal	Pin
Tare / reset button	D3 (GPIO 0) — built-in FLASH button
Status LED	D4 (GPIO 2) — built-in LED, active LOW

---

3. Open Keg Scale – Software Setup

3a. Install Arduino IDE

Download and install Arduino IDE 2.x from [arduino.cc](https://www.arduino.cc/en/software).

3b. Add ESP8266 Board Support

1. Open File → Preferences.
2. In *Additional boards manager URLs* add:

```
https://arduino.esp8266.com/stable/package_esp8266com_index.json
```

3. Open Tools → Board → Boards Manager, search esp8266, install the esp8266 by ESP8266 Community package.

3c. Select the Board

Tools → Board → ESP8266 Boards → LOLIN(WEMOS) D1 mini

Recommended upload settings:

- Upload Speed: `921600`
- CPU Frequency: `80 MHz`
- Flash Size: `4MB (FS:2MB OTA:~1019KB)`

3d. Install Libraries

Tools → Manage Libraries, search and install:

Library	Author	Minimum version
WiFiManager	tzapu	2.0.17
HX711 Arduino Library	Bogdan Necula	0.7.5

`ArduinoOTA` ships with the ESP8266 core — no separate install needed.

3e. Flash the Firmware

1. Connect the D1 Mini via USB.
2. Open `open_keg_scale/keg_scale.ino` in Arduino IDE.
3. Select the correct Port under Tools → Port.
4. Click Upload (Ctrl+U).

---

4. Open Keg Scale – First-Time Device Setup

After the first flash the device has no WiFi credentials and boots into setup mode automatically.

1. On your phone or laptop, scan for WiFi networks. Connect to `KegScale-Setup` (no password).
2. A captive portal will open automatically, or browse to `http://192.168.4.1`.
3. Fill in all fields and click Save:

Field	Description
WiFi SSID	Your 2.4 GHz network name
WiFi Password	Your network password
Plaato Server Host	IP or hostname of the machine running [open-plaato-keg](https://github.com/DarkJaeger/open-plaato-keg) (e.g. `192.168.1.50`)

Field	Description
Server Port	Default 1234
Auth Token	32-character hex token from the open-plaato-keg web UI
Calibration Factor	Leave blank on first setup; calibrate after mounting the scale
Max Keg Volume (L)	19 for a standard Cornelius / ball-lock keg
Beer Name	Displayed in the dashboard

4. The device reboots, connects to your WiFi, then connects to the Plaato server.

The LED will slow-blink (500 ms) once fully connected and reporting.

> *Re-enter setup mode at any time: Hold the FLASH button for 4+ seconds to wipe credentials, or use the web UI at* `http://<device-ip>/` *→ Reset WiFi / Enter Setup Mode.*

LED Status Reference

Pattern	Meaning
Fast blink (100 ms)	Connecting to WiFi or server
Medium blink (250 ms)	TCP connected, waiting for auth
Slow blink (500 ms)	Fully connected and reporting
3 short flashes	Tare complete
4 short flashes	Calibration saved

---

5. Open Keg Scale – Calibration

The scale ships with a calibration factor of `1.0` (uncalibrated). Calibrate once after physically mounting the load cells.

Via the server UI (recommended)

1. Remove everything from the scale platform.
2. In the open-plaato-keg web interface, click Tare.
3. Place a known-weight object (e.g. a full keg, a bag of flour, a filled water container).
4. Enter its mass in grams in the *Calibrate with known weight* field and click Send.
5. The device saves the calibration factor to flash.

Via Serial Monitor

1. Open Tools → Serial Monitor at `115200` baud.
2. Clear the platform and short-press the FLASH button to tare.
3. Place a known weight; note the raw ADC value printed.
4. Compute: `calibration_factor = raw_value / known_grams`

5. Enter the factor via the server UI calibrate command or web UI.

---

6. Open Keg Scale – OTA Updates

After the initial USB flash and WiFi join, the device advertises an Arduino OTA target.

Via Arduino IDE

1. Ensure your computer and the device are on the same network.
2. Wait for the device to connect (slow LED blink).
3. Tools → Port → choose the network port for `keg-scale-<chipid>`.
4. Click Upload. When prompted for a password enter the 32-character auth token.

Via Web UI (easiest)

Browse to `http://<device-ip>/update`, log in with:

- Username: `admin`
- Password: your 32-character auth token

Upload the compiled `.bin` file directly.

---

7. Transfer Keg Scale – Software Setup

7a. Install PlatformIO

Install [Visual Studio Code](https://code.visualstudio.com/) then add the PlatformIO IDE extension from the Extensions Marketplace.

Alternatively install the PlatformIO CLI:

```
pip install platformio
```

7b. Open the Project

Open the folder `transfer-keg-scale/transfer-keg-firmware/` in VS Code.

PlatformIO will automatically resolve all library dependencies listed in `platformio.ini` on the first build.

7c. Configure `src/main.cpp`

Edit the config block near the top of `src/main.cpp`:

```
#define WIFI_SSID "YourSSID" #define WIFI_PASSWORD "YourPassword" #define SERVER_BASE_URL
"http://192.168.1.100:3000" // open-plaato-server address #define SCALE_ID "scale1" // unique ID for this
scale #define CALIBRATION_FACTOR -430.0f // replace after running calibration sketch #define
DEFAULT_EMPTY_KG 3.8f // weight of your empty keg in kg #define DEFAULT_TARGET_KG 12.5f // weight of a
full keg in kg
```

> *Set `DEFAULT_EMPTY_KG` and `DEFAULT_TARGET_KG` for your keg type. A standard 19 L Cornelius keg weighs ~4 kg empty and ~23 kg full.*

7d. Flash the Firmware

Option A – Pre-compiled binary (quickest)

Use `esptool.py` or the [ESP8266 Flash Download Tool](https://www.espressif.com/en/support/download/other-tools) to flash `transfer-keg-scale/firmware.bin` at address `0x0`.

```
esptool.py --port COM3 write_flash 0x0 transfer-keg-scale/firmware.bin
```

Option B – Build from source (PlatformIO)

```
cd transfer-keg-scale/transfer-keg-firmware pio run -t upload
```

---

8. Transfer Keg Scale – First-Time Device Setup

1. Power the device. The OLED shows "TRANSFER SCALE" with no WiFi indicator.
2. Connect your phone or laptop to the WiFi AP `TransferScale-Setup` (password: `scaleme`).
3. Browse to `http://192.168.4.1`.
4. Fill in the setup portal:

Field	Description
WiFi SSID	Your 2.4 GHz network name
WiFi Password	Your network password
Server Base URL	Full URL to open-plaato-server (e.g. `http://192.168.1.100:3000`)
Scale ID	Unique identifier (e.g. `transfer1`)
Calibration Factor	From calibration sketch (Step 9); leave default for now
Empty Keg Weight (kg)	Weight of your empty keg
Target Weight (kg)	Weight of a full keg

5. Click Save. The device reboots, connects to WiFi, and the OLED shows [W] when connected.

> *Re-enter setup mode: Hold the FLASH button (D3) for 3+ seconds.*

OLED Display Layout

```
┌──────────────────────────┐ │ TRANSFER SCALE [W] │ ← [W] = WiFi connected │ │ │ 8.42 kg │ ← current weight
(large text) │ ████████░░ 67% │ ← fill progress bar │ │ │ Tgt:12.5kg Emp: 3.8 │ ← target / empty weights
└──────────────────────────┘
```

Progress % = `(currentWeight − emptyWeight) / (targetWeight − emptyWeight) × 100`

---

9. Transfer Keg Scale – Calibration

Via Calibration Sketch

1. In `transfer-keg-scale/transfer-keg-firmware/src/`, rename:

```
calibration.cpp.txt → calibration.cpp
```

2. In `platformio.ini`, comment out the `main.cpp` build source and uncomment `calibration.cpp` (or temporarily exclude `main.cpp` from compilation).
3. Flash and open Serial Monitor at `115200` baud:

```
pio device monitor
```

4. Follow the on-screen prompts:
   - Remove everything from the platform when asked.
   - Place a known weight (e.g. a 10 kg object) when prompted.
   - The sketch prints `CALIBRATION_FACTOR = -430.5` (example).
5. Copy that value back into the `#define CALIBRATION_FACTOR` line in `main.cpp`.
6. Undo the `platformio.ini` changes, rename `calibration.cpp` back to `calibration.cpp.txt`, and re-flash `main.cpp`.

Via Serial Commands (quick adjustment)

With `main.cpp` running, open Serial Monitor at `115200` baud:

Key	Action
`t`	Tare (zero) the scale
`+`	Increase calibration factor by 10
`-`	Decrease calibration factor by 10
`r`	Print raw HX711 reading
`c`	Print current calibration factor
`?`	Show help

Use + / – to nudge the factor until a known weight reads correctly, then save the final value to `main.cpp` and re-flash.

Calibration Notes

- If readings are negative, negate the calibration factor.
- `scale.tare()` runs at boot — keep the platform empty at power-on.
- Re-tare after ~5 minutes to account for load cell thermal warm-up.

---

10. Troubleshooting

Open Keg Scale

Symptom	Fix
No `KegScale-Setup` AP appears	Hold FLASH button 4 s to wipe credentials and restart setup
LED stuck fast-blinking	Device cannot reach WiFi or Plaato server; check credentials and server IP
Weight always 0 or garbage	Check HX711 DT→D6, SCK→D5 wiring; verify 3V3 supply
OTA port not visible in IDE	Ensure device and PC are on the same subnet; firewall may block mDNS
Volume reads wrong after calibration	Re-tare with empty platform, then re-calibrate with known weight

Transfer Keg Scale

Symptom	Fix
OLED stays blank	Check I2C address — try `0x3C` then `0x3D`; verify SDA→D2, SCL→D1
`HX711 ERROR` on boot	Check DOUT→D5, SCK→D6; confirm 5 V supply to HX711
Weight reads ~0 always	Calibration factor is wrong — re-run calibration sketch
Weight oscillates wildly	Load cell mounting issue; ensure cell corners move freely and aren't over-constrained
POST to server fails	Check `SERVER_BASE_URL`; device and server must be on the same network
WiFi never connects	Confirm 2.4 GHz SSID; D1 Mini does not support 5 GHz
No `[W]` on OLED after setup	Wrong SSID/password; hold FLASH 3 s to re-enter setup portal

